



Customer Shopping Behavior Analysis

Uncovering insights from 3,900 purchases to guide strategic business decisions

Dataset Overview

3,900

Transactions Analyzed

Purchase records across diverse product categories

18

Data Fields

Comprehensive customer and purchase details

37

Missing Values

Addressed in Review Rating column

Key features include customer demographics, purchase details, and shopping behavior metrics.

Data Preparation Process

01

Data Loading

Imported dataset using pandas for analysis

03

Missing Data Handling

Imputed missing values using median ratings by category

05

Feature Engineering

Created age groups and purchase frequency calculations

02

Initial Exploration

Checked structure with `df.info()` and summary statistics

04

Column Standardization

Rewrote columns in snake case for readability

06

Database Integration

Connected to PostgreSQL for SQL analysis

Revenue by Gender



Key Finding

Male customers generated **\$157,890** in revenue, significantly outpacing female customers at **\$75,191**.

This substantial difference suggests gender-based marketing opportunities.

Top Products by Rating

1

Gloves

Average Rating: **3.86**

2

Sandals

Average Rating: **3.84**

3

Boots

Average Rating: **3.82**

4

Hat

Average Rating: **3.80**

5

Skirt

Average Rating: **3.78**

These top-rated products represent quality offerings that could drive positive word-of-mouth.

Shipping Type Comparison

Standard Shipping

Average Purchase: **\$58.46**

Most common shipping method

Express Shipping

Average Purchase: **\$60.48**

Higher-value customers

Express shipping customers spend **\$2 more** per purchase on average, suggesting higher-value transactions.



Subscribers vs. Non-Subscribers

Subscribers (1,053 customers)

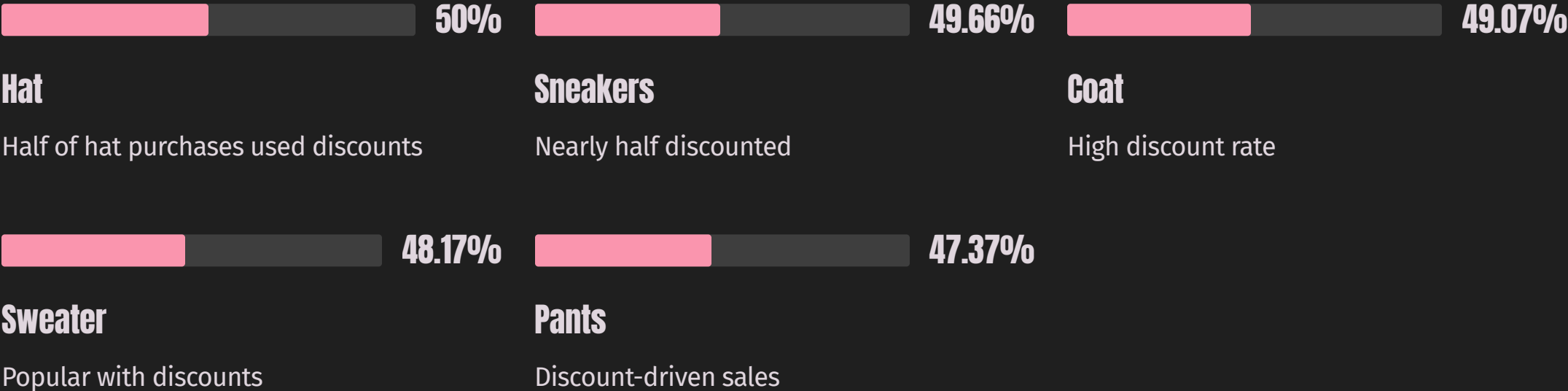
- Average spend: **\$59.49**
- Total revenue: **\$62,645**
- 27% of customer base

Non-Subscribers (2,847 customers)

- Average spend: **\$59.87**
- Total revenue: **\$170,436**
- 73% of customer base

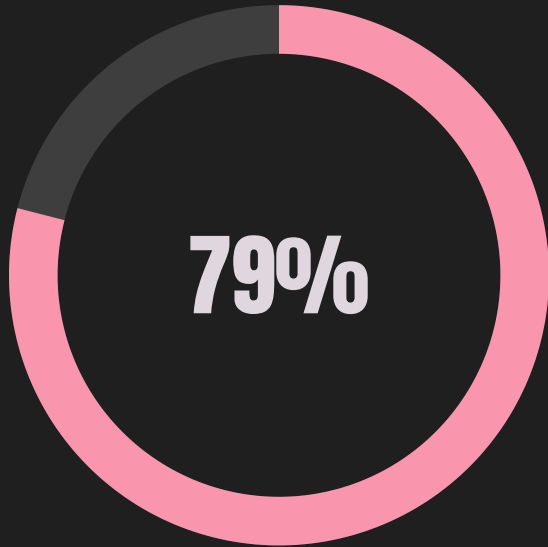
Non-subscribers generate **2.7x more revenue**, highlighting subscription program opportunity.

Discount-Dependent Products



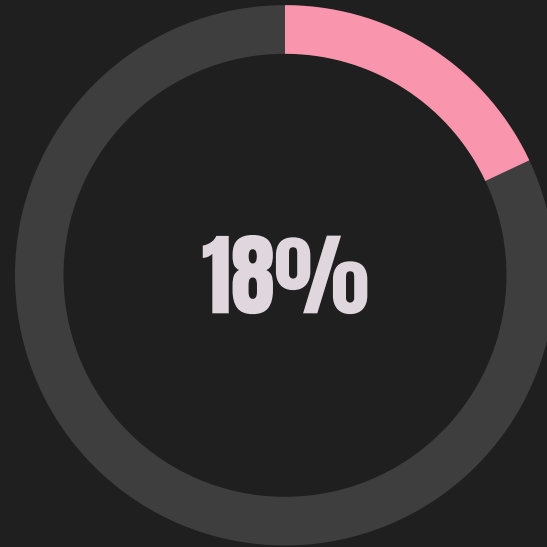
These products rely heavily on discounts, suggesting margin pressure.

Customer Segmentation



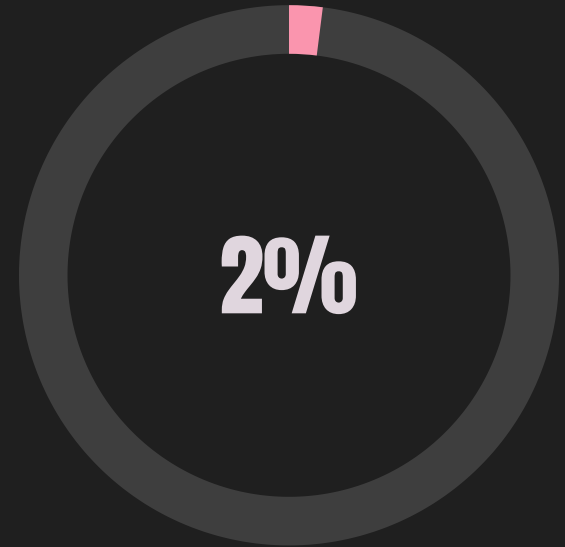
Loyal Customers

3,116 customers with consistent purchase history



Returning Customers

701 customers making second purchases



New Customers

83 first-time buyers

Strong loyalty base with 97% of customers returning or already loyal.

Strategic Recommendations



Boost Subscriptions

Promote exclusive benefits to increase subscriber base



Enhance Loyalty Programs

Reward repeat buyers to strengthen retention



Optimize Discount Policy

Balance sales growth with margin protection



Target High-Value Segments

Focus on express-shipping users and top age groups